

SE250B4 Evaluation Board Quick Start Guide

This Quick Start Guide will walk you through the basic setup required to assemble the Kit, initiate normal operation, and perform firmware upgrades.

The kit consists of three primary boards:

- SE250B4 TLB1 Board (includes SE250B4 module and RF connectors)
- SE250B4 TLB2 Board (includes touchscreen and cameras)
- SE250B4 MAIN EVB

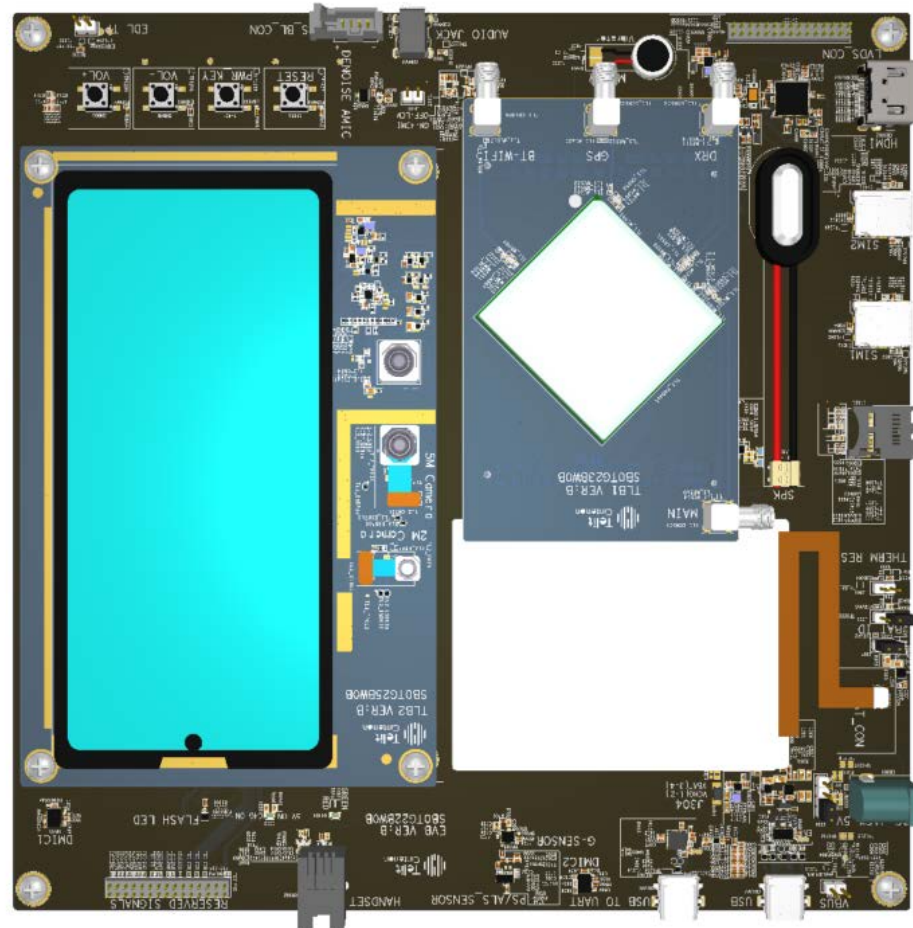


Figure 1: SE250B4 Full Kit

The TLB1 board can host the four antennas provided in the TLB2 box.

Kit Contents

- SE250B4 TLB1 Box → TLB1 board
- SE250B4 TLB2 Box → TLB2 board, DC power adapter, 1x WiFi Antenna, 2X Cellular LTE Antennas, 1x GPS Antenna, 4x screws.

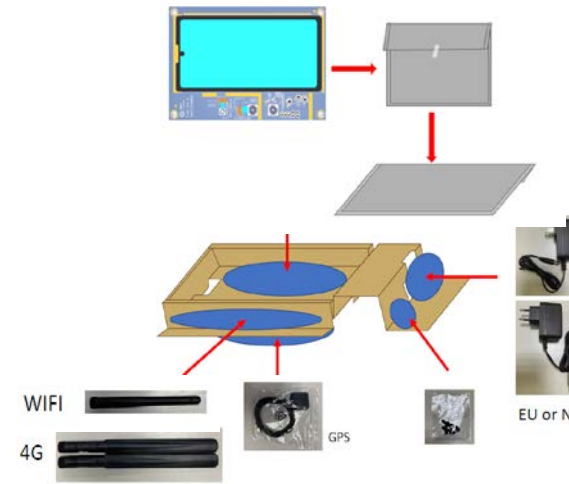


Figure 2: TLB2 Box Contents

- SE250B4 Main EVB Box → Main EVB, USB cable, spacers and screws.

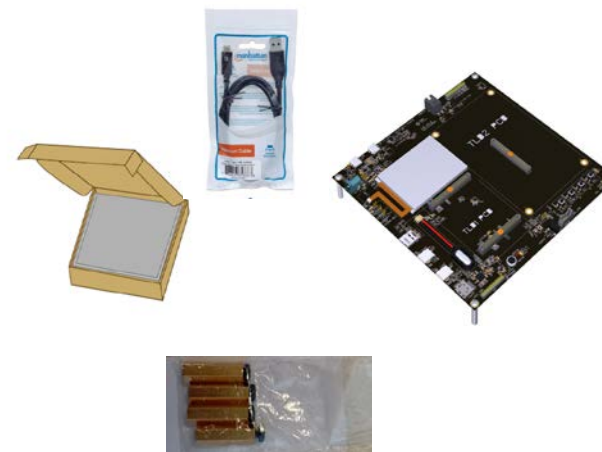


Figure 3: Main EVB Box Contents

Power Up the Unit

Before powering up the unit, check the jumper configurations:

- J304→ Selects power input circuitry configuration.
- J302→ Enable the Vbatt to the module.
- J301→ Enable the EVB-emulated BattTherm.
- J305→ Enable EVB-emulated BattID.

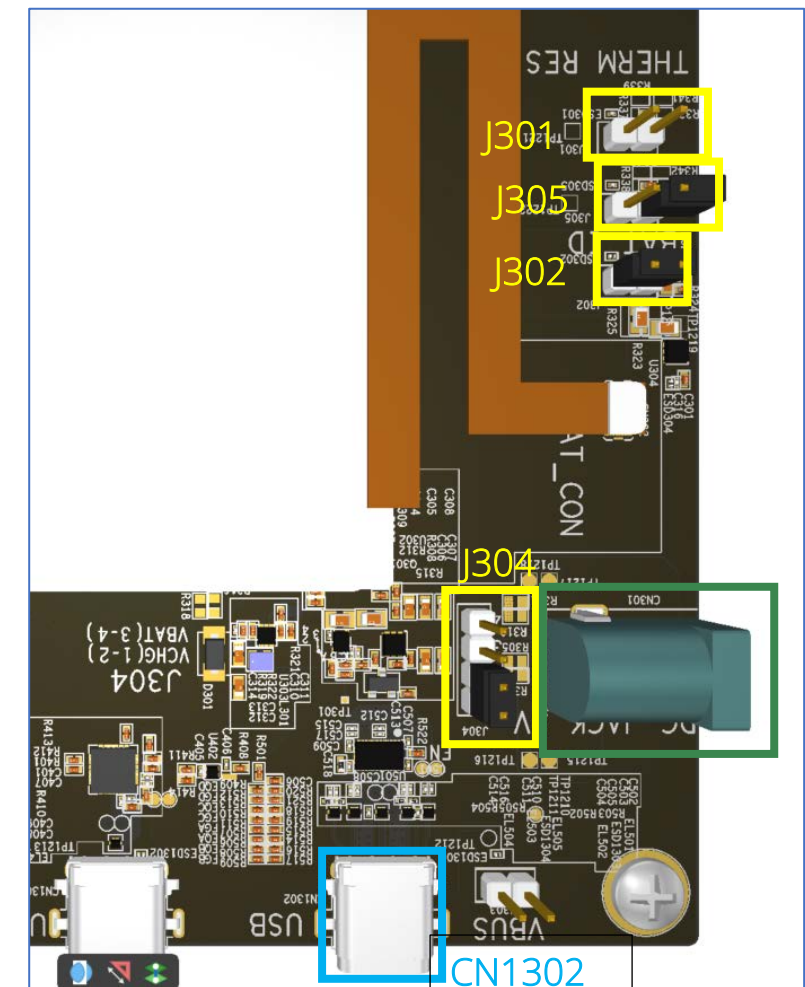


Figure 4: SE250B4 Jumpers and connectors

Battery Mode:

- Set jumper on J302. Connect a 10 W USB adapter to the CN1302 USB connector. In this mode, the PMIC inside the module will charge the battery. Once fully charged, you can disconnect the cable and use the unit as a regular Smartphone.

Non Battery Mode:

DC Barrel Jack:

- Set jumper on J302 and J304 3-4. Connect the DC power adapter to the DC barrel jack connector.
- Connect a USB host to the CN1302 USB connector for Firmware upgrades and to issue AT commands*.

USB:

- Set jumper on J302 and J303. J304 must have jumpers on pins 1-2 and 3-4. Ensure the power adapter is disconnected in this case.
- Connect a USB host to the CN1302 USB connector for power, Firmware upgrades and to issue AT commands*.

*AT port is available only in userdebug build. AT port is not available in user build FW, typically used for production. Hence it is recommended not to use AT commands, except for debug and evaluation.

1 Press Button Shortcuts

As a regular smartphone, the SE250B4 can be controlled via shortcuts for various operations:

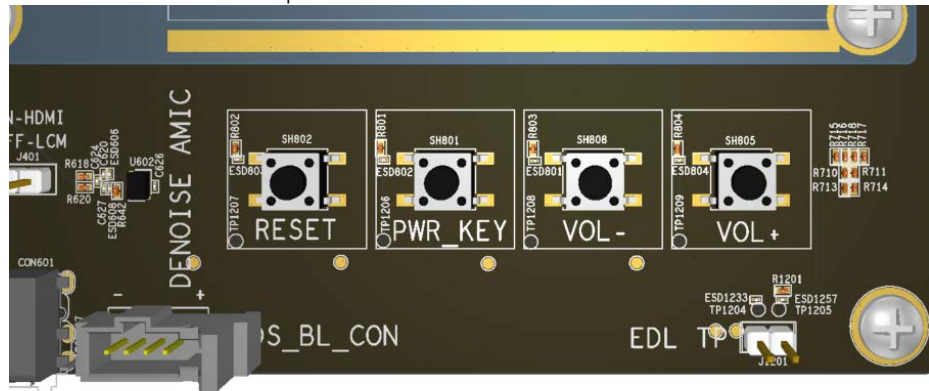


Figure 5: SE250B4 EVB Push Buttons

- **Pwr_Key** : Short press to suspend and resume; long press to trigger a reboot of DUT.
- **Reset** : Short press has no action; long press triggers a reboot of DUT.
- **Vol+** : Increases the audio volume of the Smartmodule and provides access to the audio settings menu.
- **Vol-** : Decreases the audio volume of the Smartmodule and provides access to the audio settings menu.
- **"PWKEY" + "VOL+"**: Press to access to power-up menu.

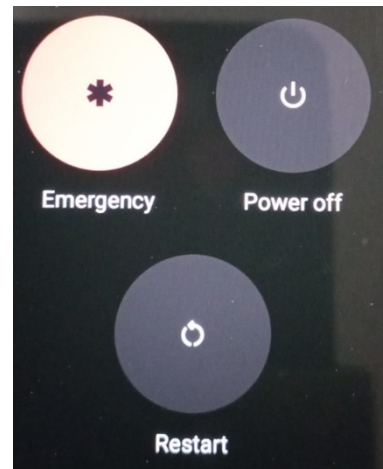


Figure 6: SE250B4 Power-Up Menu

- **"PWKEY" + "VOL-"**: Press to take a screenshot.

2 AT Commands and ADB Shell

AT commands*

1. Power up the module.
2. Connect the USB cable provided in the package to the CN1302 connector.
3. Open ELTIMA or any other Terminal application like teraterm and select the COM port with a baud rate of 115200.

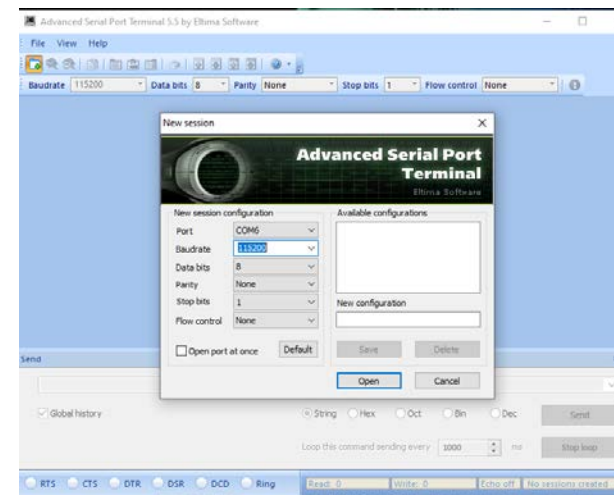


Figure 7: Terminal COM Port Connection

Now, you can issue the standard AT commands of QCM platforms for debugging and testing. For a comprehensive list of available AT commands on this platform, refer to the QCM document, [80-vr432-1-j-comprehensive-at-command_set.pdf](#).

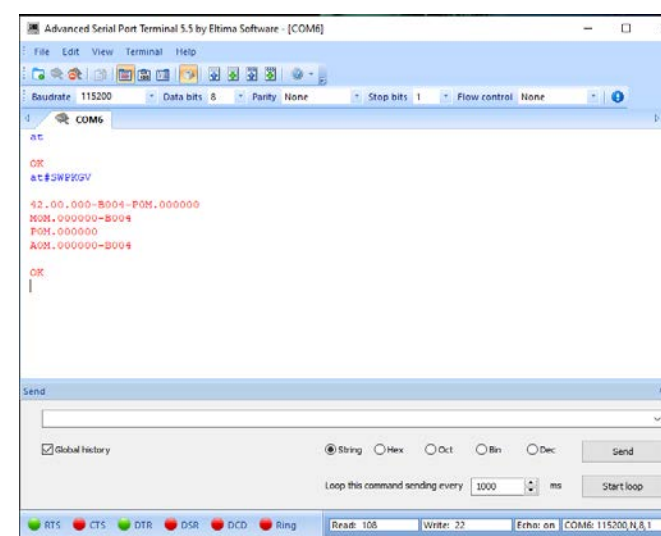


Figure 8: Terminal AT Command

ADB Shell Commands and APK Installation

1. To obtain the ADB-platform-tools, go to <https://developer.android.com/studio/releases/platform-tools>. Download the r3.10.4-windows version, and extract the files.

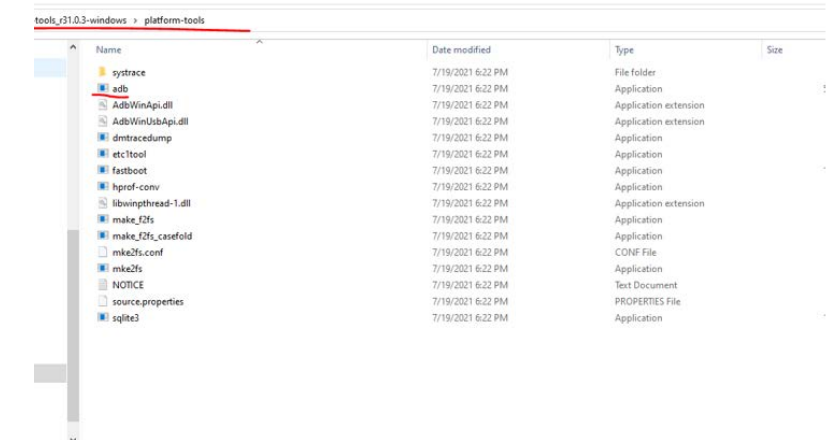


Figure 9: ADB tutorial

2. Set the extracted folder as a new environment variable on your PC.

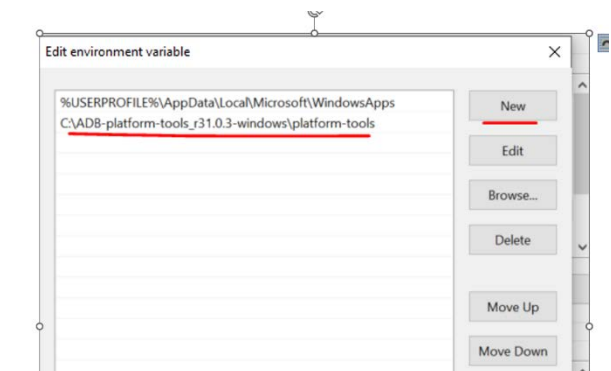


Figure 10: ENV variable

Connect the device to your PC to access to ADB shell. Open a command prompt and type "adb devices" to view the list of attached devices. The list of attached devices will be shown as below:

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You're now ready to go!

3 Firmware Upgrading

1. Go to <https://www.telit.com/support-tools/download-zone/>
2. Under SE250B4, download the SE250B4 USB drivers and the appropriate FW version for your SKU.
3. Decompress and install USB Drivers.
4. In normal operation, connect the system to your PC via a USB cable and run the TFI application.
5. The application will automatically update the firmware, and the system will reboot.

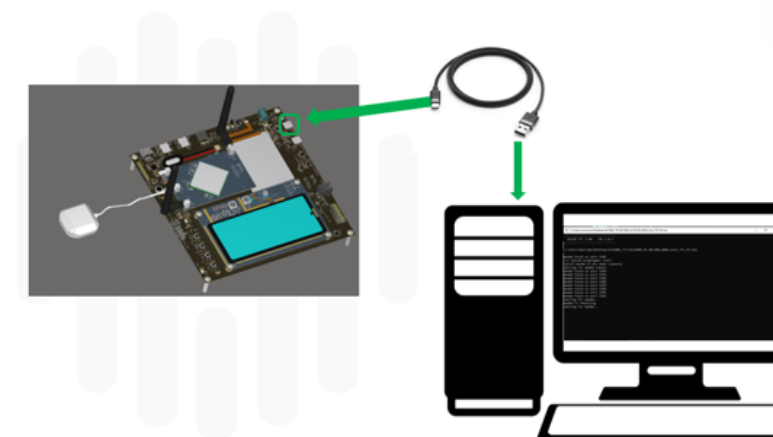
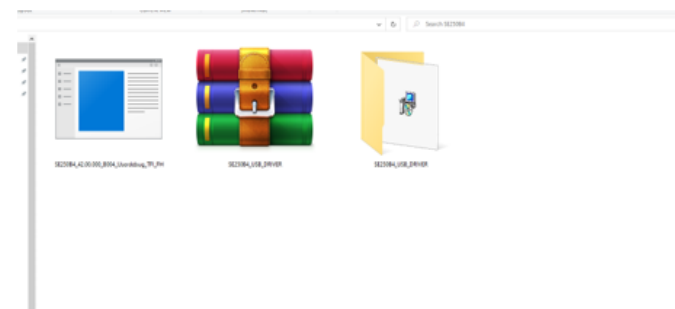


Figure 13: Firmware Application Example

```

Microsoft Windows [Version 10.0.19045.3208]
(c) Microsoft Corporation. All rights reserved.

C:\Users\marcomo>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
c56897a0    device

C:\Users\marcomo>

```

Figure 11: Command Prompt

To install an APK, run the following command after downloading the required files from any APK website:

```
adb install -r -t name_of_your_app.apk
```

For example, to install the sensor APK, use:

```
adb install -r -t SensorApp2.8.apk
```

```

Command Prompt

C:\APKAPPS>adb install -r -t SensorApp2.8.apk
Performing Streamed Install
Success
C:\APKAPPS>

```

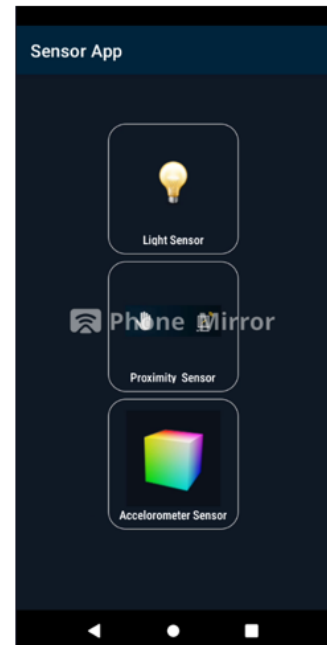
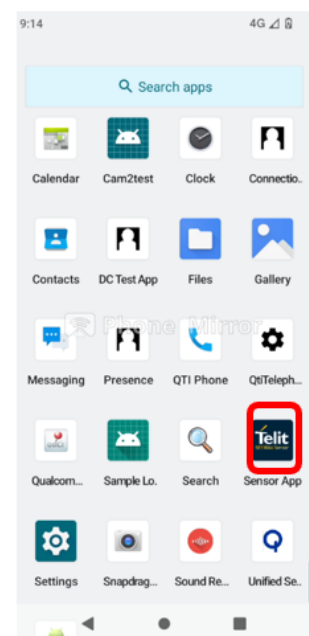


Figure 12: APK Installation and Run

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